Quantum Scattering Theory: Born Approximation

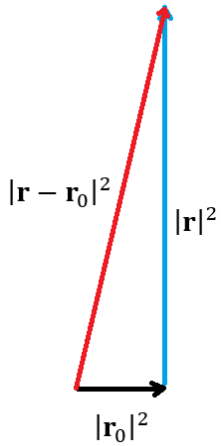
In our integral form of the Schrodinger equation

$$\psi(\mathbf{r}) = \psi_0(\mathbf{r}) - \frac{m}{2\pi\hbar^2}\int\left(\frac{e^{ik|\mathbf{r}-\mathbf{r}_0|}}{|\mathbf{r}-\mathbf{r}_0|}\right)V(\mathbf{r}_0)\psi(\mathbf{r}_0)d^3\mathbf{r}_0$$

We have the wavefunction on both sides of the equal sign (green), which are the same function, which means we need to know something about the solution to find the solution

We therefore need the Born approximation method

We have our potential $V(\mathbf{r}_0)$ centered around $\mathbf{r}_0$ and we want to calculate $\psi(\mathbf{r})$ at points far from $\mathbf{r}_0$, i.e., when $|\mathbf{r}| \gg |\mathbf{r}_0|$ for all points in the above equation



We then do

$$|\mathbf{r}-\mathbf{r}_0|^2 = r^2 + r_0^2 - 2\mathbf{r}\cdot\mathbf{r}_0$$

Because $r^2 \gg r_0^2$ we can ignore the r_0^2 term

$$|\mathbf{r}-\mathbf{r}_0|^2 \approx r^2\left(1 - 2\frac{\mathbf{r}\cdot\mathbf{r}_0}{r^2}\right)$$

Take square root of both sides

$$|\mathbf{r}-\mathbf{r}_0| \approx r\sqrt{1 - 2\frac{\mathbf{r}\cdot\mathbf{r}_0}{r^2}}$$

Use binomial expansion to get

$$|\mathbf{r}-\mathbf{r}_0| \approx r\left(1 - \frac{\mathbf{r}\cdot\mathbf{r}_0}{r^2}\right)$$

Distribute the r

$$|\mathbf{r}-\mathbf{r}_0| \approx r - \frac{\mathbf{r}\cdot\mathbf{r}_0}{r}$$

Then with $\frac{\mathbf{r}}{r} = \hat{r}$ (the r-direction) we get

$$|\mathbf{r}-\mathbf{r}_0| \approx r - \hat{r}\cdot\mathbf{r}_0$$

Then we set k for the incident wave (in the z-direction) to be

$$\mathbf{k}' = k\hat{z}$$

And k for the scattered wave (in the r-direction) to be

$$\mathbf{k} = k\hat{r}$$

Which means that

$$e^{ik|\mathbf{r}-\mathbf{r}_0|} \approx e^{ikr}e^{-i\mathbf{k}\cdot\mathbf{r}_0}$$

And so, our integral $\psi(\mathbf{r}) = \psi_0(\mathbf{r}) - \frac{m}{2\pi\hbar^2}\int\left(\frac{e^{ik|\mathbf{r}-\mathbf{r}_0|}}{|\mathbf{r}-\mathbf{r}_0|}\right)\big(V(\mathbf{r}_0)\psi(\mathbf{r}_0)\big)d^3\mathbf{r}_0$ becomes

$$\psi(\mathbf{r}) = \psi_0(\mathbf{r}) - \frac{m}{2\pi\hbar^2}\int\left(\frac{e^{ikr}e^{-i\mathbf{k}\cdot\mathbf{r}_0}}{r-\hat{r}\cdot\mathbf{r}_0}\right)\big(V(\mathbf{r}_0)\psi(\mathbf{r}_0)\big)d^3\mathbf{r}_0$$

Since $r \gg \mathbf{r}_0$ we can do

$$\psi(\mathbf{r}) = \psi_0(\mathbf{r}) - \frac{m}{2\pi\hbar^2}\int\left(\frac{e^{ikr}e^{-i\mathbf{k}\cdot\mathbf{r}_0}}{r}\right)\big(V(\mathbf{r}_0)\psi(\mathbf{r}_0)\big)d^3\mathbf{r}_0$$

$$\psi(\mathbf{r}) = \psi_0(\mathbf{r}) - \frac{m}{2\pi\hbar^2}\int\left(\frac{e^{ikr}}{r}e^{-i\mathbf{k}\cdot\mathbf{r}_0}\right)\big(V(\mathbf{r}_0)\psi(\mathbf{r}_0)\big)d^3\mathbf{r}_0$$

Where $\frac{e^{ikr}}{r}$ is a spherical wave which is not dependent on $\mathbf{r}_0$

Recall that our incident wavefunction is $\psi_0(\mathbf{r}) = Ae^{ikz}$ which means we get

$$\psi(\mathbf{r}) = Ae^{ikz} - \frac{e^{ikr}}{r}\frac{m}{2\pi\hbar^2}\int\big(e^{-i\mathbf{k}\cdot\mathbf{r}_0}\big)\big(V(\mathbf{r}_0)\psi(\mathbf{r}_0)\big)d^3\mathbf{r}_0$$

Where $\frac{e^{ikr}}{r}\frac{m}{2\pi\hbar^2}$ is a spherical wave multiplied by a constant which we can then put as $\frac{m}{A2\pi\hbar^2}$ and write

$$A\frac{e^{ikr}}{r}$$

Which gives us

$$\psi(\mathbf{r}) = Ae^{ikz} - A\frac{e^{ikr}}{r}\int\big(e^{-i\mathbf{k}\cdot\mathbf{r}_0}\big)\big(V(\mathbf{r}_0)\psi(\mathbf{r}_0)\big)d^3\mathbf{r}_0$$

Then put $f(\theta) = \int\big(e^{-i\mathbf{k}\cdot\mathbf{r}_0}\big)\big(V(\mathbf{r}_0)\psi(\mathbf{r}_0)\big)d^3\mathbf{r}_0$ as our scattering factor to get

$$\psi(\mathbf{r}) = Ae^{ikz} - f(\theta)A\frac{e^{ikr}}{r}$$

Meaning that

$$f(\theta,\phi) = -\frac{m}{2\pi\hbar^2}\int e^{-i\mathbf{k}\cdot\mathbf{r}_0}V(\mathbf{r}_0)\psi(\mathbf{r}_0)d\mathbf{r}_0$$

The Born approximation assumes weak scattering, and so the incident wave is not significantly altered by the potential, i.e.,

$$\psi(\mathbf{r}_0) \approx \psi_0(\mathbf{r}_0)$$

Which is (using $\mathbf{k}' = k\hat{z}$)

$$\psi_0(\mathbf{r}_0) = Ae^{ikz_0} = Ae^{i\mathbf{k}'\cdot\mathbf{r}_0}$$

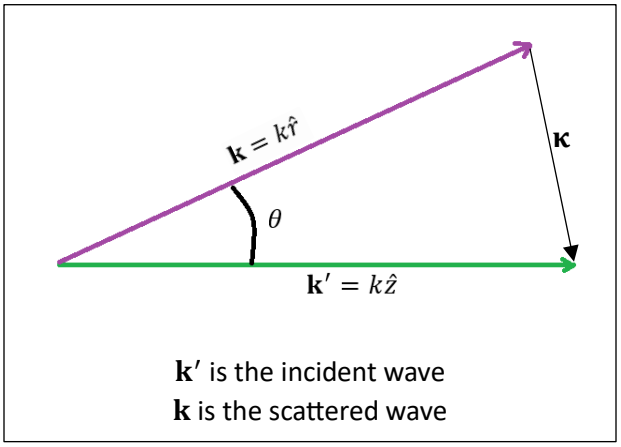
And so we get for $f(\theta,\phi) = \frac{m}{2\pi\hbar^2}\int e^{-i\mathbf{k}\cdot\mathbf{r}_0}V(\mathbf{r}_0)\psi(\mathbf{r}_0)d\mathbf{r}_0$ (after replacing $\psi(\mathbf{r}_0)$ with the above)

$$f(\theta,\phi) = -\frac{m}{2\pi\hbar^2}\int e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}_0}V(\mathbf{r}_0)\psi(\mathbf{r}_0)d\mathbf{r}_0$$

For low energy we will have $(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}_0 \approx 0$ and so

$$f(\theta,\phi) \approx -\frac{m}{2\pi\hbar^2}\int V(\mathbf{r}_0)\psi(\mathbf{r}_0)d\mathbf{r}_0$$

Which is the scattering factor from the Born approximation



$\mathbf{k}'$ is the incident wave
$\mathbf{k}$ is the scattered wave

At higher energy we have

$$(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}_0 = \boldsymbol{\kappa}\cdot\mathbf{r}_0\cos\theta_0$$

Meaning that z-axis, i.e., the polar axis for $\mathbf{r}_0$, is lying along $\boldsymbol{\kappa}$

This gives us the scattering factor of

$$f(\theta) = -\frac{m}{2\pi\hbar^2}\int e^{i\boldsymbol{\kappa}\cdot\mathbf{r}_0\cos\theta_0}V(\mathbf{r}_0)\mathbf{r}_0^2\sin\theta_0\,d\theta_0 d\phi_0$$

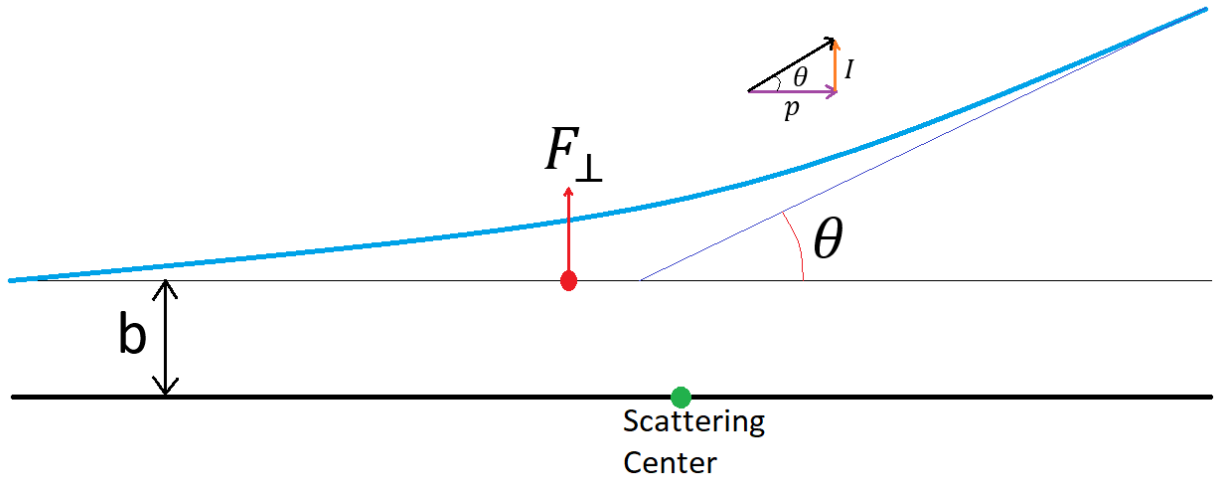
Since the scattering potential is assumed to be spherically symmetrical, then it is only a function of θ and so we can integrate out the ϕ_0 to get 2π which then gives

$$f(\theta) = -\frac{2m}{\hbar^2\kappa}\int_0^\infty rV(r)\sin(\kappa r)\,dr$$

Where $\kappa = 2k\sin\frac{\theta}{2}$ and carries the angular dependence of $f(\theta)$

In classical physics we have the transverse impulse given by a force acting transverse to the path of the particle for some amount of time

$$I = \int F_{\perp}\, dt$$



Where the angle is then the inverse tangent of the impulse divided by the incident momentum p

$$\theta = \tan^{-1}\left(\frac{I}{p}\right)$$

This is a first order impulse approximation (zeroth order is no deflection at all)

Similarly, the zeroth order Born approximation is no affect on incident plane wave, which was

$$f(\theta) \approx -\frac{m}{2\pi\hbar^2}\int V(\mathbf{r}_0)\psi(\mathbf{r}_0)d\mathbf{r}_0$$

And the first order was

$$f(\theta) \approx -\frac{2m}{\hbar^2\kappa}\int_0^{\infty} rV(r)\sin(\kappa r)\, dr$$

Which came from the integral form of the Schrodinger equation which has the form

$$\psi(\mathbf{r}) = \psi_0(\mathbf{r}) + \int G(\mathbf{r}-\mathbf{r}_0)V(\mathbf{r}_0)\psi(\mathbf{r}_0)d\mathbf{r}_0 \qquad g(\mathbf{r}) = \frac{m}{2\pi\hbar^2}\frac{e^{ikr}}{r}$$

With $\psi_0(\mathbf{r})$ the incident wave

This schematically has the form

$$\psi = \psi_0 + \int gV\psi$$

But then if we plug $\psi = \psi_0 + \int gV\psi$ into the ψ in the integral sign we get

$$\psi = \psi_0 + \int gV\left[\psi_0 + \int gV\psi\right]$$

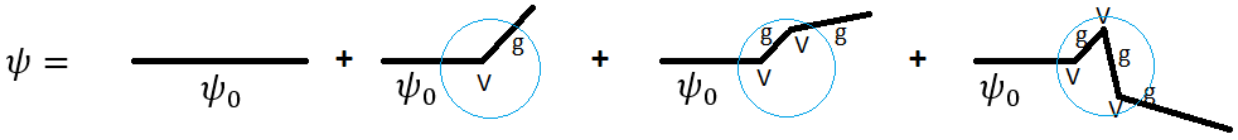
Which is

$$\psi = \psi_0 + \int gV\psi_0 + \int\int gV\psi$$

Then do it again to get

$$\psi = \psi_0 + \int gV\psi_0 + \int\int gVgV\psi_0 + \int\int\int gVgVgV\psi$$

To keep getting higher order terms, which give more and more complicated interactions with the scattering potential



(Similar to Feynman diagrams, with each higher order being a more complex interaction)

We can see that g (the Green's function) is a **response** to some interaction with the scattering potential V